# CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

**OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM**

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## Experiment 12: Spring Boot Data JPA with Database Repository

Course: Advanced Java Programming Laboratory [23CS3504]
Category: Experiments / Practicals | Experiment #12 | Academic Year: 2026-27

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### 1. Overview & Objective:

Persist entity models using Spring Data JPA JpaRepository and custom query methods.

### 2. Practical Tasks & Verification Steps:

1. Annotate Student entity class with @Entity, @Table, @Id, and @GeneratedValue strategy = GenerationType.IDENTITY
2. Create StudentRepository interface extending JpaRepository<Student, Long>.
3. Define custom Spring Data query method findByBranchAndSemester String branch, int semester .
4. Verify automatic schema creation with spring.jpa.hibernate.ddl-auto=update and H2/MySQL database.

**Submission Guidelines & Verification:**

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